

Study note: from a ring transform to the sub-cubic sum

GMaster, 24 September 2026

Every identity below is the one implemented in `gmaster/_sht/dc.py`.
Nothing is assumed from the v2 march except the transform it is replacing.

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1 The transform, before any algorithm

A scalar map on the sphere has spherical-harmonic coefficients

$$a_{\ell m} = \int_{S^2} f(\theta, \varphi) Y_{\ell m}(\theta, \varphi)^* d\Omega, \quad (1.1)$$

and the inverse is $f = \sum_{\ell m} a_{\ell m} Y_{\ell m}$. The spherical harmonics separate,

$$Y_{\ell m}(\theta, \varphi) = \lambda_{\ell m}(\cos \theta) \frac{e^{im\varphi}}{\sqrt{2\pi}},$$

with $\lambda_{\ell m}$ the orthonormal associated Legendre function on $[-1, 1]$ (the Condon–Shortley phase is the one shared with ducc0; it rides along and never changes the argument below). The area element is $d\Omega = \sin \theta d\theta d\varphi$.

HEALPix, at resolution N_{side} , places its pixels on rings of constant θ . On ring r the azimuthal samples are equally spaced, so the φ integral on that ring is a discrete Fourier transform. After that Fourier transform, what remains of (1.1) at one fixed order m is a sum over rings only:

$$a_{\ell m} = \sum_r \lambda_{\ell m}(x_r) F_m(\theta_r) w_r, \quad x_r = \cos \theta_r. \quad (1.2)$$

$F_m(\theta_r)$ is the Fourier coefficient of the ring, and w_r is the quadrature weight of that ring (pixel area, and the usual HEALPix ring weight). Synthesis is the transpose: given $a_{\ell m}$, form

$$F_m(\theta_r) = \sum_{\ell \geq |m|} \lambda_{\ell m}(x_r) a_{\ell m} \quad (1.3)$$

and Fourier-synthesise each ring.

Equations (1.2) and (1.3) are the whole latitudinal problem. The azimuthal FFT is not discussed again. The bandlimit is $L = \ell_{\text{max}} + 1$, with $\ell_{\text{max}} = 3N_{\text{side}} - 1$ in every timing below, so there are $O(L)$ orders and $O(L)$ degrees in each.

2 What the march does, and what it costs

The associated Legendre functions in ℓ , at fixed m and x , obey a three-term recurrence. The v2 march evaluates that recurrence once per ring: for every pair (ℓ, m) it produces $\lambda_{\ell m}(x_r)$, then dots with the ring coefficient. The number of pairs is $\sum_m (L - |m|) \sim L^2/2$, and each is repeated on $N_{\text{ring}} \sim 4N_{\text{side}} \sim L$ rings, so the work is

$$O(L^2 N_{\text{ring}}) = O(L^3).$$

That count is the definition of the march. The rest of this note never forms $\lambda_{\ell m}(x_r)$.

3 One order is a polynomial in $\cos^2 \theta$

Fix $m \geq 0$. The functions $\lambda_{\ell m}(x)$ are polynomials in x of degree ℓ , times a fixed power of $\sqrt{1 - x^2}$ that depends only on m . They have a parity:

$$\lambda_{\ell m}(-x) = (-1)^{\ell+m} \lambda_{\ell m}(x).$$

Split the sum (1.3) into even and odd steps in ℓ relative to m . Write $p \in \{0, 1\}$ for that parity, and keep only the degrees

$$\ell = m + p + 2k, \quad k = 0, 1, 2, \dots$$

On those degrees $\lambda_{\ell m}(x)$ is an even or odd function of x times a fixed factor that does not depend on k . Set

$$y = x^2 = \cos^2 \theta \in [0, 1].$$

Then the k -dependence sits in a polynomial in y . Define the orthonormal versions of those polynomials by folding the Legendre row,

$$\phi_k(y) = \psi_{m+p+2k}(\sqrt{y}), \quad (3.1)$$

where ψ_ℓ is $\lambda_{\ell m}$ with the m -dependent factor divided out and the result renormalised so that $\{\phi_k\}$ is orthonormal on $[0, 1]$ with its weight. Orthonormality is the only property used from here:

$$\int_0^1 \phi_j(y) \phi_k(y) d\mu(y) = \delta_{jk}. \quad (3.2)$$

The northern and southern hemispheres are the same polynomials in y , with a sign $(-1)^\ell$ between them. One (m, p) problem therefore covers both hemispheres. There are four such problems for each pair $\pm m$ (two signs of m , two parities); negative m is the complex conjugate of positive m for a real map, so it is not a new polynomial.

A synthesis at this (m, p) is now exactly

$$f(y_r) = \sum_{k=0}^{n-1} c_k \phi_k(y_r), \quad (3.3)$$

with c_k a rescaling of $a_{m+p+2k, m}$ and n the number of degrees of this parity, $n \sim (L - m)/2$. Equation (3.3) is the object to evaluate. The march would build each $\phi_k(y_r)$ by recurrence. The next sections replace that.

4 Multiplication by y is a tridiagonal matrix

Any three consecutive orthonormal polynomials obey a three-term recurrence. The reason is degree, not a special property of Legendre. The product $y \phi_k$ is a polynomial of degree $k + 1$ in the same family, so

$$y \phi_k = \sum_{j=0}^{k+1} J_{jk} \phi_j.$$

Orthonormality gives $J_{jk} = \int \phi_j y \phi_k d\mu$. The integrand is symmetric in (j, k) , so J is symmetric. If $|j - k| > 1$ then $y \phi_k$ has degree at most $k + 1 < j$, and ϕ_j is orthogonal to every polynomial of lower degree, so $J_{jk} = 0$. Only the three diagonals survive:

$$y \phi_k = E_k \phi_{k+1} + D_k \phi_k + E_{k-1} \phi_{k-1}. \quad (4.1)$$

$E_k > 0$ by the standard sign convention on leading coefficients, and D_k is real. In matrix form, on the coefficient vector of $(\phi_0, \dots, \phi_{n-1})$, multiplication by y is the $n \times n$ real symmetric tridiagonal Jacobi matrix T with diagonal D_k and off-diagonal E_k .

5 The nodes are the eigenvalues

Let $T = VYV^\top$ be the spectral decomposition: V orthogonal, $Y = \text{diag}(y_0, \dots, y_{n-1})$. The eigenvalues y_j are simple and lie in $(0, 1)$. They are the zeros of ϕ_n . That is Golub–Welsch: $\phi_n(y_j) = 0$, and the j -th column of V , normalised to unit Euclidean length, is the vector of polynomial values at that zero, scaled by the Gauss quadrature weight w_j of the node,

$$V_{kj} = \sqrt{w_j} \phi_k(y_j), \quad w_j = \left(\sum_{k=0}^{n-1} \phi_k(y_j)^2 \right)^{-1}. \quad (5.1)$$

(The weight in (5.1) is the Christoffel number. It is not the HEALPix ring weight w_r of (1.2).) Equation (5.1) is the bridge from the matrix V , which we will apply without forming ϕ_k , back to the polynomial values.

6 Christoffel–Darboux, derived

Proposition 6.1. *For the orthonormal recurrence (4.1),*

$$(z - y) \sum_{k=0}^{n-1} \phi_k(z) \phi_k(y) = E_{n-1} (\phi_n(z) \phi_{n-1}(y) - \phi_{n-1}(z) \phi_n(y)). \quad (6.1)$$

Proof. Apply (4.1) at z and at y :

$$\begin{aligned} z \phi_k(z) &= E_k \phi_{k+1}(z) + D_k \phi_k(z) + E_{k-1} \phi_{k-1}(z), \\ y \phi_k(y) &= E_k \phi_{k+1}(y) + D_k \phi_k(y) + E_{k-1} \phi_{k-1}(y). \end{aligned}$$

Multiply the first by $\phi_k(y)$ and the second by $\phi_k(z)$, and subtract. The D_k terms cancel, and what remains is a difference of neighbours:

$$\begin{aligned} (z - y) \phi_k(z) \phi_k(y) &= E_k (\phi_{k+1}(z) \phi_k(y) - \phi_k(z) \phi_{k+1}(y)) \\ &\quad - E_{k-1} (\phi_k(z) \phi_{k-1}(y) - \phi_{k-1}(z) \phi_k(y)). \end{aligned}$$

Sum from $k = 0$ to $k = n - 1$. The sum on the right telescopes. The $k = 0$ end vanishes because there is no ϕ_{-1} . The only surviving boundary term is the one at $k = n - 1$, with coefficient E_{n-1} , which is (6.1). $\square$

Now specialise to a node. By §5, $\phi_n(y_j) = 0$, so (6.1) at $y = y_j$ collapses to

$$\sum_{k=0}^{n-1} \phi_k(z) \phi_k(y_j) = E_{n-1} \frac{\phi_n(z) \phi_{n-1}(y_j)}{z - y_j}. \quad (6.2)$$

Take the polynomial expansion (3.3) and insert the spectral expansion of the coefficients. Since V is orthogonal, $c = Vw$ means $w = V^\top c$ and $c_k = \sum_j V_{kj} w_j$. Therefore

$$f(z) = \sum_{k=0}^{n-1} c_k \phi_k(z) = \sum_{j=0}^{n-1} w_j \sum_{k=0}^{n-1} V_{kj} \phi_k(z). \quad (6.3)$$

Substitute $V_{kj} = \sqrt{w_j} \phi_k(y_j)$ from (5.1) into the inner sum:

$$\sum_{k=0}^{n-1} V_{kj} \phi_k(z) = \sqrt{w_j} \sum_{k=0}^{n-1} \phi_k(y_j) \phi_k(z) = \sqrt{w_j} E_{n-1} \frac{\phi_n(z) \phi_{n-1}(y_j)}{z - y_j},$$

where the second equality is (6.2). The same identity (5.1) at degree $n - 1$ says $V_{n-1,j} = \sqrt{w_j} \phi_{n-1}(y_j)$, so $\sqrt{w_j} \phi_{n-1}(y_j) = V_{n-1,j}$. The inner sum is therefore

$$\sum_k V_{kj} \phi_k(z) = E_{n-1} \phi_n(z) \frac{V_{n-1,j}}{z - y_j}.$$

Put that into (6.3) and restore w_j as the j -th entry of $V^\top c$:

$$\boxed{f(z) = E_{n-1} \phi_n(z) \sum_{j=0}^{n-1} \frac{V_{n-1,j} (V^\top c)_j}{z - y_j}}. \quad (6.4)$$

This holds at every z that is not a node y_j . A HEALPix ring $y_r = \cos^2 \theta_r$ is such a point, except for the size-1 problem, where a node can sit on a ring; that case is §9.

Read (6.4) from left to right as an algorithm:

1. $w = V^\top c$. This is a linear map on the n coefficients of one order. It does not look at the rings.
2. For each ring $z = y_r$, the sum $\sum_j w_j V_{n-1,j} / (y_r - y_j)$. This is a Cauchy sum from the n nodes to that ring. The numbers $V_{n-1,j}$ are the last row of V , a property of T alone.
3. Multiply by the scalar $E_{n-1} \phi_n(y_r)$, a property of the ring and of (m, p) , independent of the map.

No value of $\phi_k(y_r)$ for $k < n$ was computed. Analysis is the Euclidean adjoint. The scalar in step 3 is real, the Cauchy kernel $1/(y_r - y_j)$ is symmetric in the two sets of points, and V is orthogonal, so the adjoint is: multiply the ring residual by $E_{n-1} \phi_n(y_r)$, apply the transposed Cauchy sum, then apply V . Same plan, reversed order, transposed kernels.

7 Applying $V^\top$ without storing V

Forming V would be $O(n^3)$ for one order and $O(L^4)$ for the sphere, which is worse than the march. V is the eigenvector matrix of a tridiagonal matrix, and a tridiagonal eigenproblem splits.

Tear T into two principal blocks T_1, T_2 of half the size, and one coupling $\rho = E_{n/2}$ across the cut. After moving that coupling onto a rank-one spike,

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix} + \rho uu^\top \quad (7.1)$$

for a vector u that is nonzero in only two entries (the two corners that were joined). This is Cuppen's splitting. Suppose the two blocks are already diagonalised, $\text{diag}(T_1, T_2) = Q \text{diag}(d) Q^\top$. Then T is a diagonal matrix plus a rank-one update, in the Q basis. Its eigenvalues λ are the roots of the secular equation

$$1 + \rho \sum_i \frac{u_i^2}{d_i - \lambda} = 0, \quad (7.2)$$

one root in each gap between the d_i (and one outside each end). The corresponding unit eigenvector has components proportional to $u_i/(d_i - \lambda)$. So the eigenvector matrix of the merge, in the basis of the children's eigenvectors, has entries

$$U_{ij} = \frac{z_i c_j}{d_i - \lambda_j}. \quad (7.3)$$

Gu and Eisenstat's form replaces the raw u_i by a computed z_i so that (7.3) is the exact eigenvector matrix of a tridiagonal matrix within a few units in the last place of T . That is what makes the merge stable; a raw $u_i/(d_i - \lambda)$ is not, once two poles nearly coincide.

Applying U or $U^\top$ to a vector is the Cauchy sum (7.3). It is the same shape as the ring sum in (6.4). The divide-and-conquer tree is: diagonalise the 16×16 leaves directly, then at each level apply one Cauchy merge per block. A leaf eigenvector block is not stored. Its 16 eigenvalues determine it, and a half-warp rebuilds the 16×16 block from those eigenvalues by a twisted factorisation of the torn tridiagonal. The tridiagonal entries are closed form in (m, ℓ) from (4.1). Stored cost of a leaf is 16 eigenvalues, not 16^2 eigenvectors: 4 bytes per coefficient instead of 64.

There are $O(\log n)$ levels. A direct Cauchy sum at a level of size n is $O(n^2)$ per block and $O(n)$ blocks, hence $O(n^2)$ per level, and $O(n^2 \log n)$ per order. That would still be $O(L^3 \log L)$ over all orders, no better than the march. The next section is what removes the extra n .

8 The Cauchy sums are one-dimensional fast multipole sums

Both (6.4) and (7.3) are sums

$$g(x) = \sum_j \frac{q_j}{x - y_j}$$

with charges q_j on a set of real points y_j and targets x on the real line. If a target interval and a source interval are separated by at least their own length, then for every source y in the source interval and every target x in the target interval the kernel has a uniformly convergent expansion in $(x - c)^{-1}$ about the source centre c ,

$$\frac{1}{x - y} = \sum_{p=0}^{P-1} \frac{(y - c)^p}{(x - c)^{p+1}} + \text{remainder}.$$

The sum over a well-separated source interval collapses to P moments $\sum_j q_j (y_j - c)^p$, independent of the target, and each target evaluates a P -term polynomial in $1/(x - c)$. Near interactions, where the intervals touch, stay direct. With P fixed, a binary tree of this splitting evaluates the sum in

$O(N)$ or $O(N \log N)$ work for N sources and N targets. The implementation uses $P = 12$ and leaves of 32 poles, and it uses the direct sum when a merge has fewer than 256 poles. That is a parameter choice, not a new identity: any fixed P leaves the complexity $O(n \log n)$ per order.

An order of length n therefore costs $O(n \log n)$ to apply $V^\top$, and $O(n)$ or $O(n \log n)$ for the final node-to-ring Cauchy sum. Summing over parities and over $m = 0, \dots, L-1$, with $n \sim (L-m)/2$, gives

$$\sum_{m=0}^{L-1} O((L-m) \log(L-m)) = O(L^2 \log L)$$

for one scalar transform. The march was $O(L^3)$.

9 When a denominator vanishes

Two corrections are required because (6.4) and (7.3) divide by a difference of real numbers.

Adjacent eigenvalues of the two children in (7.1) can lie closer together than a float64 unit in the last place of their values. Their stored difference is then noise. The plan stores each adjacent gap exactly (as a float32 correction on top of the double-float poles). The hot loop uses a clamped reciprocal, and each target replaces at most three nearest terms by those exact values afterwards. The replacement is branch-free. Deflating the pair by a Givens rotation is the wrong fix here: the secular root that sits between the two poles is closer than the perturbation the rotation introduces, and the resulting error is of order 10^{-3} .

Separately, a node y_j can land on a ring y_r . The size-1 problem does this exactly. At $N_{\text{side}} = 4096$, spin 2, $m = 179$, the closest approach on the board is 8.9×10^{-15} . The plan lists every node-ring pair closer than 10^{-9} , with the float64 value of $y_r - y_j$. If the two coincide, the summand has a limit, from differentiating (6.2) or equivalently from the normalisation (5.1),

$$E_{n-1} \phi'_n(y_j) = \frac{\phi_0(y_j)}{V_{0j} V_{n-1,j}}.$$

A small kernel writes those terms in place of the reciprocal.

10 Spin 2 is the same tridiagonal problem

For a spin- s field the latitudinal functions are Wigner d -functions. With $a = m + s$, $b = |m - s|$, $x = \cos \theta$ and $n = \ell - \max(m, s)$,

$$d_{m,-s}^\ell(\theta) = \left(\sin \frac{\theta}{2}\right)^a \left(\cos \frac{\theta}{2}\right)^b P_n^{(a,b)}(x). \quad (10.1)$$

(The phase and the power of two that convert this to the Condon–Shortley normalisation are independent of the summation index inside one order; they scale c_k and do not change T .) $P_n^{(a,b)}$ is a Jacobi polynomial. Jacobi polynomials satisfy a three-term recurrence in n because they are orthogonal polynomials, by the same degree argument as §4. Written on the orthonormal functions $u_\ell = \sqrt{(2\ell+1)/2} d_{m,-s}^\ell$, that recurrence is

$$x u_\ell = a_{\ell+1} u_{\ell+1} + b_\ell u_\ell + a_\ell u_{\ell-1}, \quad (10.2)$$

with the explicit coefficients

$$b_\ell = -\frac{ms}{\ell(\ell+1)}, \quad (10.3)$$

$$a_\ell = \frac{\sqrt{(\ell^2 - m^2)(\ell^2 - s^2)}}{\ell \sqrt{4\ell^2 - 1}}. \quad (10.4)$$

Equation (10.3) is the only structural difference from spin 0. For $s = 0$ one has $b_\ell = 0$, the matrix is off-diagonal, even and odd ℓ decouple, and §3 folds them into $y = x^2$. For $s \neq 0$ the diagonal is nonzero, so there is one tridiagonal per m , of length $L - \max(m, s)$, with no parity split, and the nodes are Gauss–Jacobi nodes in $x = \cos \theta$ rather than in $y = \cos^2 \theta$. The ring sum runs over all rings, not over a hemisphere.

From (10.2) onward nothing changes. T is still real symmetric tridiagonal, (6.1)–(6.4) never used $D_k = 0$, and Cuppen’s splitting (7.1) never used it either. Negative orders do not need a second plan:

$$d_{-m,-s}^\ell(\theta) = (-1)^{\ell+s} d_{m,-s}^\ell(\pi - \theta). \quad (10.5)$$

The right-hand side is the same vector of polynomial values on the rings read in reverse order, times a sign that depends on ℓ but not on θ . It is a second right-hand side of the same traversal.

The coefficients (10.3) and (10.4) were checked against a direct sympy expansion of the Jacobi recurrence to 10^{-16} . The check is a comparison of closed forms, not a fit.

11 The iteration never returns to $a_{\ell m}$

Write (6.4) as a matrix. Let C be the linear map that sends the node-space vector w to the rings by the Cauchy sum and the scalar $E_{n-1}\phi_n$, and let f be the per-degree normalisation that converts $a_{\ell m}$ into the orthonormal coefficients c . Synthesis and analysis of one spin are

$$S = C V^\top \text{diag}(f), \quad A = \text{diag}(f) V C^\top. \quad (11.1)$$

For the normalised transforms of either spin, $f^2 = 1/(2\pi)$. (For spin s the bare normalisation and the factor $\sqrt{(2\ell+1)/2}$ in u_ℓ multiply to this same constant; the mirror sign in (10.5) squares to 1.)

MASTER refinement at n_{iter} iterations is the map

$$a \leftarrow a - A(Sa - \text{map}), \quad (11.2)$$

applied n_{iter} times, starting from $a = 0$ or from a previous estimate. Substitute (11.1). Set $w = V^\top(a/f)$, i.e. $a = f V w$, which is step 1 of §6 done once. Then

$$Sa = C V^\top \text{diag}(f) (f V w) = C V^\top (f^2) V w.$$

V is orthogonal and $f^2 = 1/(2\pi)$ is a scalar, so $V^\top (f^2) V = (1/(2\pi))I$ and

$$Sa = \frac{1}{2\pi} C w.$$

The correction $A(Sa - \text{map})$ pushed back to w -space is

$$V^\top \text{diag}(f)^{-1} A(Sa - \text{map}) = V^\top C^\top (Sa - \text{map}) = C^\top \left(\frac{Cw}{2\pi} - \text{map} \right),$$

where the first equality uses $A = \text{diag}(f) V C^\top$ and $\text{diag}(f)^{-1} \text{diag}(f) = I$. Therefore (11.2) in the w coordinate is

$$\boxed{w \leftarrow w - C^\top \left(\frac{Cw}{2\pi} - \text{map} \right)} \quad (11.3)$$

and $a = f V w$ is formed only after the last iteration. One iteration is two applications of C and a ring-space residual. The tree that applies V and $V^\top$, which is about 70% of the cost of a transform, runs once per refinement loop instead of once per S and once per A . With $n_{\text{iter}} = 3$ that is once instead of $2 \cdot 3 + 1 = 7$ times.

The residual $Cw/(2\pi) - \text{map}$ is a ring map. Its azimuthal structure is already a Fourier series (the output of C is $F_m(\theta_r)$). Folding that series, $\text{FFT}(\text{IFFT}(F))$, is the periodic sum that the ring transform needs, and it does not require a second FFT inside the iteration. The loop stays in node space and in ring-Fourier space.

12 Cost, memory, and where the engine stops

Work per scalar transform: $O(L^2 \log L)$, from the last display in §8. Measured on one NVIDIA RTX PRO 6000 Blackwell (96 GB), doubling L from $N_{\text{side}} = 2048$ to 4096 multiplies a D&C pass by about 4.2 and a march pass by 6 to 16. The log-log slope in N_{side} over that interval is 2.05 / 2.06 (spin 0, synthesis / analysis) and 2.09 / 2.07 (spin 2). An $O(N^2 \log N)$ model predicts 2.13 on that interval; $O(N^3)$ predicts 3.

The plan (eigenvalues, Gu–Eisenstat z , adjacent gaps, the last row of each V , the scalars $E\phi_n$) is geometry only. It is $O(L^2 \log L)$ numbers. The march’s recurrence tables are $O(L^2)$. At $N_{\text{side}} = 4096$ the accounted device peak is 37.5 GiB for spin 0 and 59.9 GiB for spin 2, against 25.0 and 43.9 GiB for the march. Above $L = 12288$, which is $N_{\text{side}} = 8192$ at $\ell_{\text{max}} = 3N_{\text{side}} - 1$, the router does not build the plan: the host build would take hours and about 55 GiB per spin. At that size the latitudinal step falls back to the march, and the memory wall that remains is the azimuthal chirp-Z, which is outside this note.

On the same card, spin 0, warm medians, synthesis / analysis at $n_{\text{iter}} = 0$ / analysis at $n_{\text{iter}} = 3$, in milliseconds:

N_{side}	D&C	v2 march	$n_{\text{iter}} = 3$ ratio
1024	5.9 / 6.7 / 15	6.9 / 7.3 / 43	2.8×
2048	24 / 29 / 57	37 / 42 / 251	4.4×
4096	123 / 145 / 250	511 / 453 / 3110	12×

The error against `ducc0` on the same HEALPix operator stays between 2×10^{-6} and 3×10^{-5} at every size on that board. The march, whose error is the float32 recurrence, grows to about 10^{-4} at $N_{\text{side}} = 4096$.

13 What was not used

The derivation used orthonormality, the three-term recurrence, the spectral theorem for a real symmetric tridiagonal matrix, and one telescoping sum. It did not use a fast cosine transform, a precomputed Legendre table, or the compensated float32 arithmetic of the march. The march remains the algorithm below $L = 3072$, where the tree’s launch cost is larger than the $O(L^3)$ recurrence, and above $L = 12288$, where the plan does not fit the build.